

Quick Check

Lesson 2: Meaning of Growth and Development; Structure of Monocot and Dicot Seeds

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 2

A Grade 11 class is discussing how growth differs from development in a maize plant. Which statement correctly distinguishes the two?

- A** Growth and development mean the same thing, because both describe a plant getting taller.
- B** Growth is a permanent increase in size, dry mass or cell number, while development is a change in form and complexity, such as cell differentiation.
- C** Growth is the formation of new organs such as leaves and flowers, while development is only an increase in height.
- D** Growth is any increase in size, even a temporary one, while development is a decrease in size.

A farmer watches her maize plants over several weeks. Which observation is an example of development rather than only growth?

- A** The maize plant forms a tassel and cobs.
- B** The maize root extends deeper into the soil.
- C** The maize stem becomes taller.
- D** The maize stem becomes thicker.

A group splits open a soaked bean seed and finds a small embryo between the two cotyledons. Which part of the embryo will grow into the root of the new plant?

A The plumule

B The radicle

C The cotyledon

D The testa

A learner cuts a soaked maize grain lengthwise and puts a drop of iodine solution on the cut surface. The large whitish area (endosperm) turns blue-black. What does this show?

- A** The endosperm is dead tissue.
- B** The endosperm contains starch, so it is the food store of the maize grain.
- C** The endosperm contains the embryo of the maize grain.
- D** The endosperm has absorbed water, so it will become the root.

In a maize grain, the coleoptile is a sheath around the plumule. What is its job during germination?

- A** It protects the plumule as it pushes up through the soil.
- B** It absorbs digested food from the endosperm.
- C** It protects the radicle as the root emerges.
- D** It allows water to enter through a small pore.

Which row correctly compares a maize grain (monocot) with a bean seed (dicot)?

- A** Maize: one cotyledon, no food store. Bean: two cotyledons, food stored in the endosperm.
- B** Maize: one cotyledon (scutellum), food stored in the endosperm. Bean: two cotyledons, food stored in the cotyledons.
- C** Maize and bean: two cotyledons, food stored in the endosperm.
- D** Maize: two cotyledons, food stored in the cotyledons. Bean: one cotyledon, food stored in the endosperm.

A learner draws a model of how a maize grain starts to grow after planting. Which arrow path correctly shows how stored food reaches the embryo?

A Endosperm, then the scutellum, then the embryo

B Testa, then the embryo

C Cotyledons, then the embryo

D Coleoptile, then the radicle

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).